

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

19th May 2023

Subject SP2 – Life Insurance

Time allowed: 3 Hours 15 Minutes (10.15 – 13.30 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
3. *Attempt all questions beginning your answer to each question on a separate sheet.*
4. *Mark allocations are shown in brackets.*
5. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

Q. 1) You have appeared for an interview with a Company ABC that has been selling conventional with-profits business for past 20 years. How would you respond to the following questions:

i) What are the different types of bonuses that a company can pay under with-profits contracts? (7)

ii) Define how the asset share of a with-profit insurance contract may be determined. (7)

You have been selected by the Company. The Company has been selling 3 types of regular premium participating products:

Product A: Under this product, the company declared a bonus of 4% p.a. under the simple reversionary bonus for the first 4 years, 6% p.a. for the next 6 years and 5% p.a. for the next 5 years and 7% p.a. for the last 5 years.

Product B: Under this product, the company declared compound reversionary bonus of 5% p.a. for the first 8 years, 6% p.a. for the next 5 years and 4% p.a. for the last 7 years.

Product C: Under the super-compound bonus approach, it declared the sum assured rate as 4% p.a. throughout the period and the attaching bonus rate as 8% p.a. throughout the period.

iii) Calculate and show the total attaching bonus under each of the products considering a sum assured of INR 100 for a 20-year term. Show your workings clearly at each point at which there is a bonus rate change. (5)

You have been asked to develop a model to calculate the asset share which can assist the Company in bonus declaration and help in determination of policy payouts.

iv) Explain how the asset share helps in determination of reversionary and terminal bonus rates to policyholders. (7)

v) Suggest possible reasons why the payout from a with-profits policy might be different from the policy's asset share. (8)

vi) How can an asset share be negative and what would be the consequences to the company? (2)

[36]

Q. 2) Please provide adequate justification for the purchase of following products:

a) Mrs Adrija, a 30-year-old single mother to a 5-year daughter purchases a 25-year term assurance product.

b) Mrs Binita, a 45-year-old female purchases a single premium unit linked policy. It is a whole life policy with considerable premium. The death benefit is equal to the premium.

c) Mrs Chetna, a 65-year-old female purchases an index-linked immediate annuity with a guaranteed payment period of five years.

[6]

- Q. 3)** XYZ Life Insurance Company is planning to launch a Without profit whole life insurance policy with a built in option. You have been asked to calculate the additional annual premium payable for the option for the following sample model point:

The policy shall have an option at the 10th policy anniversary such that the policyholder may take out a further whole life assurance policy for a sum assured of INR 20,000 at the company's standard premium rates and without undergoing medical underwriting.

Sum Assured : INR 20,000

Current Age : 50

Gender : Male

Standard mortality: AM92 Select

Interest : 4% per annum

Expenses : None

Proportion exercising option : 30% of all policyholders

Mortality after exercising : AM92 Ultimate mortality with ten years added to actual age

Basic premiums are paid annually in advance throughout life, under all policies. The additional premium is payable until the option date or until earlier death. Benefits are paid at the end of the year of death.

[8]

Q. 4)

- i) The insurance company is reviewing its pricing assumptions, describe how the insurance company would determine its renewal expense assumption for future model. (4)
- ii) An insurance company has just reviewed its pricing assumptions. Following the review, the company revised the annual renewal expense pricing assumption on one of its products to INR 50 per policy. The average annual renewal expense for this product over recent years has been INR 70 per policy.

Discuss possible reasons why the annual renewal expense assumption has been set to be less than the average annual renewal expense over recent years.

(7)

[11]

Q. 5)

- i) Describe the general features of an internal unit-linked fund. (3)
- ii) A life insurance company issues annual premium unit linked endowment assurance policies. The death benefit under the policy is equal to unit fund value or sum assured whichever is higher. The maturity value is the value of the units at the maturity date.

On each policy anniversary during the first ten years, the policyholder has an option to increase the sum assured by 10% of its then current value without evidence of health. The sum assured can not be increased after the tenth policy anniversary. The premium will increase in same proportion with increase in sum assured.

A policy can be surrendered at any time prior to maturity. The surrender value is equal to the value of units held at the date of surrender and a surrender penalty is applied if the surrender is made in the first 5 years of the policy.

The insurer company will review the level of premium at the tenth policy anniversary and every five years thereafter. A policy charge equal to a percentage of the fund is deducted from the policy value each month. The rate of mortality assumed are guaranteed under the contract for the first ten years, but can be reviewed by the company every five years thereafter.

Comment on the following product design factors:

- a) Marketability of product. (4)
 - b) The financing requirements of contract. (4)
 - c) The main risks the insurance company is exposed to under this product. (8)
 - d) Describe the main risks the policyholder is exposed to under this product. (4)
- [23]**

Q. 6)

- i) List the four climate related events which have resulted climate risk for life insurance industry. (1)
 - ii) List the main four stakeholders which are responsible for managing climate risk. (1)
 - iii) Discuss the actions that should be taken by these stakeholders to manage the climate risk. (6)
- [8]**

Q. 7)

Describe the Value at Risk approach used to assess a risk-based solvency capital requirement for a life insurance company. **[8]**
